Problem 6

Quantum computation

: Variational quantum algorithms with symmetry constraints

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Term

- # observable

usually represented by a matrix

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \rightarrow \underset{\text{Measure Eigenvalues}}{\{\ \lambda_1, \lambda_2, \lambda_3\ \}}$$

Eigenvalues of this matrix tell us what possible values the variable can take when we measure it in an experiment

- # hamiltonian ⟷ Energy

most important observables in physics

associated to the ENERGY

Hamiltonian tells us what possible values of the energy can measure in that system

Eigenvalues ⟷ Energy Levels

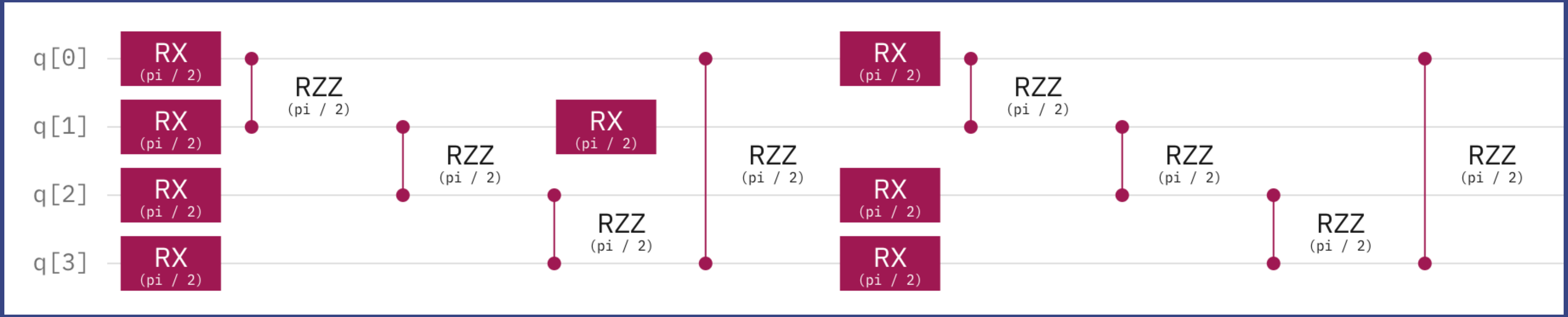
warm-up #1

Show that the output state of the following circuit always obey the global Z2 symmetry:

$$|\psi(\theta,\phi)\rangle = \prod_{j=1}^{p}\left[\prod_{i=1}^{N} e^{-iZ_iZ_{i+1}\theta_{ji}} \prod_{i=1}^{N} e^{-iX_i\phi_{ji}}\right]|\psi_0\rangle$$

$$e^{-iZ_iZ_{i+1}\theta} \rightarrow RZZ(2\theta)$$

$$e^{-iX\theta} \rightarrow RX(2\theta)$$



warm-up #2

Implement the above circuit using PennyLane and find the ground state of the following Hamiltonian using variational optimization:

$$H = -\sum_i Z_i Z_{i+1}$$

$$|GHZ\rangle_N = \frac{1}{\sqrt{2}}\left[|0\rangle^N + |1\rangle^N\right]$$

or 'cat state' ?!

n-qubit cat state $|🐱_n\rangle = \frac{1}{\sqrt{2}}(|0^n\rangle + |1^n\rangle)$.

$$E_0 \leq E_1 \leq E_2$$

" ground state "

lowest eigenvalue that is physical systems have a **lowest possible energy** which is known as the **'ground energy'**

Watts, A. B., Kothari, R., Schaeffer, L., & Tal, A. (2019, June). Exponential separation between shallow quantum circuits and unbounded fan-in shallow classical circuits. In Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (pp. 515-526).

warm-up #2

$$\boldsymbol{H} = -\sum_i \boldsymbol{Z_i Z_{i+1}} \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \rightarrow$$

eigenvector

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = -1 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

eigenvalue

$|0\rangle \rightarrow 1$

$|1\rangle \rightarrow -1$

$$minimize\ H \Rightarrow maximize \sum_i Z_i Z_{i+1} \rightarrow \sum_i Z_i Z_{i+1} \quad Z_1 Z_2 \quad Z_2 Z_3 \quad \cdots$$

$+1 \times +1$

-1×-1

$\sum_i Z_i Z_{i+1}$	$Z_1 Z_2$	$Z_2 Z_3$
$\lvert 0\rangle \ \lvert 0\rangle$	$\lvert 0\rangle \ \lvert 0\rangle$	$\lvert 0\rangle \ \lvert 0\rangle$
$-\lvert 1\rangle \ -\lvert 1\rangle$	$\lvert 1\rangle \ \lvert 1\rangle$	$\lvert 1\rangle \ \lvert 1\rangle$

$|0........0\rangle$

$|1........1\rangle$

$a|0........0\rangle + b|1........1\rangle$

$-(a+b)$

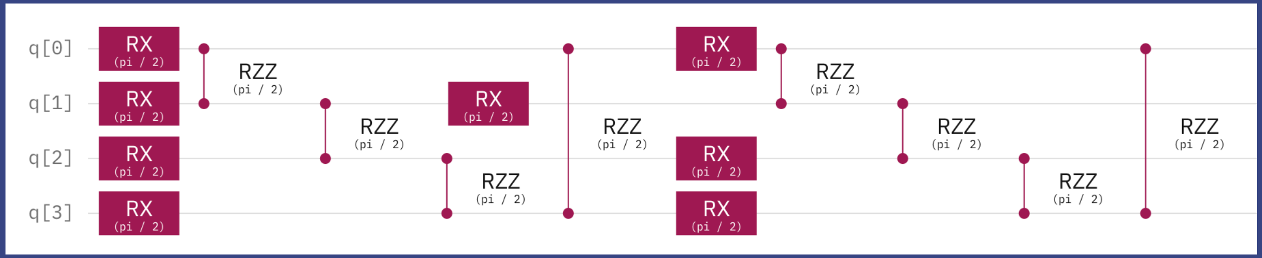
$maximize\ 'a+b'$

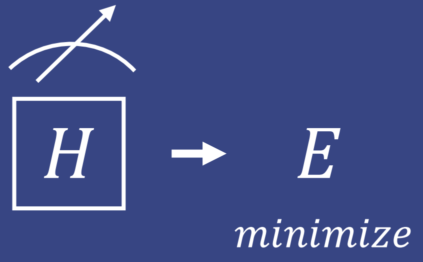
$|a|^2 + |b|^2 = 1$

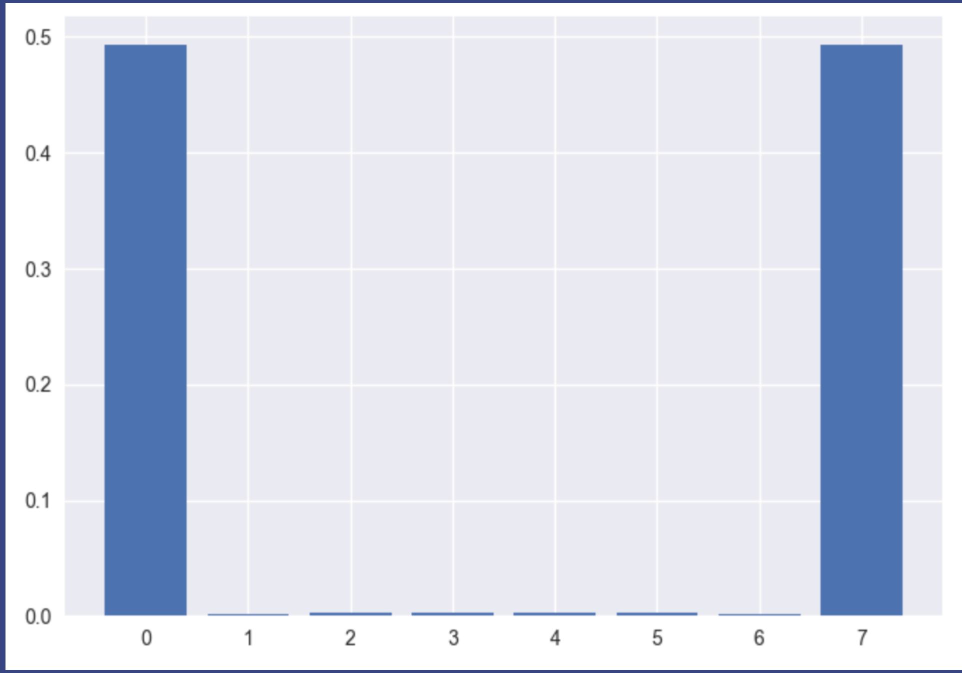
$a = \frac{1}{\sqrt{2}} \quad b = \frac{1}{\sqrt{2}}$

$$\rightarrow \quad \boldsymbol{|GHZ\rangle_N = \frac{1}{\sqrt{2}}[|0\rangle^N + |1\rangle^N]}$$

warm-up #2







⬟ warm-up #3 (bonus)

< Proof by Contradiction >

Let's prove that *"GHZ cannot be made to constant depth"*

Assume that

only finite input, output, unitary are available

If it is possible to input and output at finite size,

It is meaningless because it uses the entire $Unitary$

that sends $initial\ state$ to GHZ state

arbitrary finite input and output of $Unitary$

→ can implement by connecting 2 input, 2 output $Unitary$ to constant depth

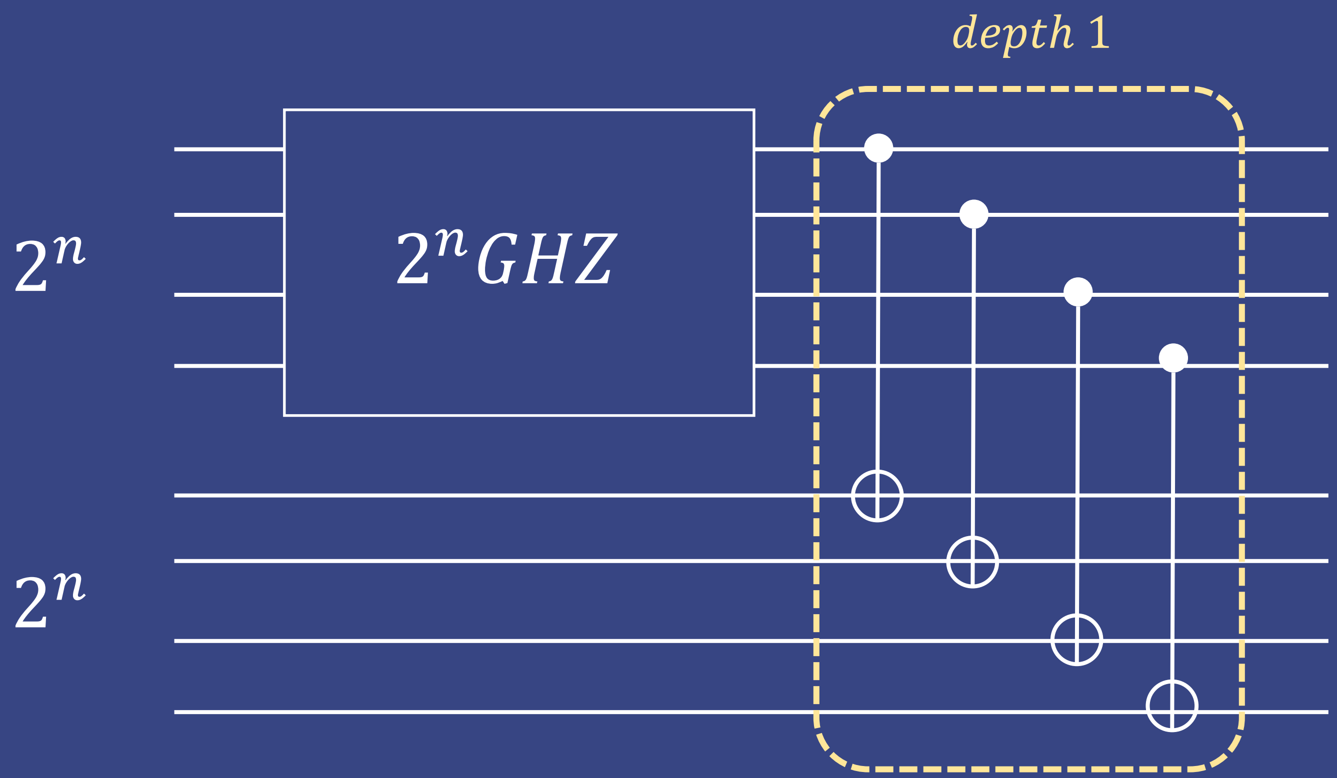
(by Quantum Computation and Quantum Information)

So let's consider using only gates below $N\ qubits$ and $2\ qubits$

⬟ warm-up #3 (bonus)

Using $CNOT$ gate well, it is easy to prepare at $logN$ depth

(increase the size by 2 times like the bottom)



$\rightarrow \quad 2^{n+1} GHZ$

$\therefore$ Easy at $logN$ depth

Is it possible to make more efficient (constant depth)?

⬟ warm-up #3 (bonus)

$\textit{entanglement}$ is interaction

$\rightarrow 2\ qubit\ gate$

⬇

BUT

$3\ qubit\ \cdots\ 1\ layer\ 1\ CNOT$

$4\ qubit\ \cdots\ 1\ layer\ 2\ CNOT$

$\Longrightarrow$ circuit depth is large as required entanglement

$\Longrightarrow$ cannot create an n-GHZ gate with constant depth

problem #1

Find single and two-qubit gates with the global U(1) symmetry.

$$Q(\theta) = exp\left[i\theta \sum_{i=1}^{N} Z_i\right]$$

1 qubit

$$\begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} \rightarrow \begin{array}{c} a \times I \\ |a|^2 = 1 \end{array}$$

2 qubit

$$\begin{bmatrix} e^{2i\theta} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{-2i\theta} \end{bmatrix} \rightarrow$$

2 qubit (cont.)

problem #1

< 2 qubit (cont.)>

$$Q(\theta) = exp\left[i\theta\sum_{i=1}^{N} Z_i\right]$$

$$N = 2 \rightarrow Q(\theta) = \begin{bmatrix} e^{2i\theta} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{-2i\theta} \end{bmatrix}$$

$$Q^\dagger U Q = U \Leftrightarrow UQ = QU$$

$$let\ U = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}$$

$$U = \begin{bmatrix} a & 0 & 0 & d \\ 0 & f & g & 0 \\ 0 & j & k & 0 \\ m & 0 & 0 & p \end{bmatrix}$$

$$U^{-1} = U^\dagger \qquad \begin{bmatrix} a & d \\ m & p \end{bmatrix} \quad \begin{bmatrix} f & g \\ j & k \end{bmatrix}$$

It can be confirmed that this is
equivalent to the condition that each is ‘Unitary'

problem #1

$$SWAP, \qquad R_Z, \qquad CZ$$

Double Excitation

$$|0011\rangle \rightarrow \cos\frac{\phi}{2}\,|0011\rangle + \sin\frac{\phi}{2}|0011\rangle$$

$$|1100\rangle \rightarrow \cos\frac{\phi}{2}\,|1100\rangle - \sin\frac{\phi}{2}|0011\rangle$$

Single Excitation

$$U(\phi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\frac{\phi}{2} & -\sin\frac{\phi}{2} & 0 \\ 0 & \sin\frac{\phi}{2} & \cos\frac{\phi}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

problem #2

Consider the second quantized molecular Hamiltonian for N2 in the sto-3g basis. Obtain its qubit basis Hamiltonian (as the sum of Pauli strings) using OpenFermion and PennyLane. Show that the resulting Hamiltonian has the U(1) symmetry.

hamiltonian = Energy

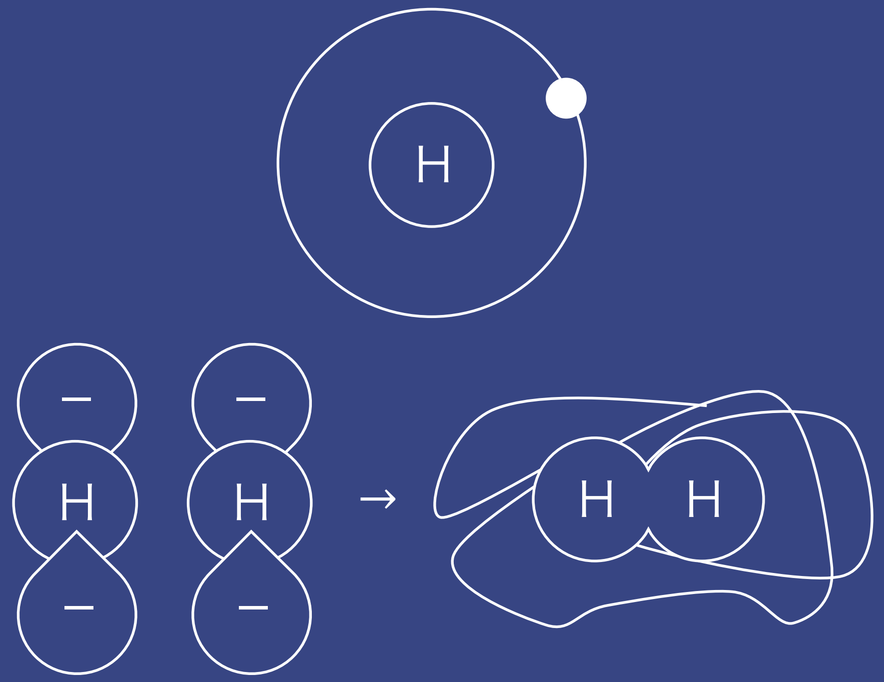
Find the best electron placement to minimize energy

second quantized

$$orbital\ n \leftrightarrow qubit\ 2n$$

expression of electron placement as quantum state

+) additional explanation



problem #2

$$XR_z(\phi) = R_z(-\phi)X$$
$$YR_z(\phi) = R_z(-\phi)Y$$
$$ZR_z(\phi) = R_z(\phi)Z$$

The number of X and Y must be even

(or sum of the number of X and Y must be even)

```python
# read file Hamiltonian20.txt
H20 = open("Hamiltonian20.txt", "r")
H20 = H20.read()
H20 = H20.split("\n")
# extract the elements in [ ]
H20 = [i.split("[")[1].split("]")[0] for i in H20]
sum = [(i.count("X") + i.count("Y")) %2==0 for i in H20]
if all(sum):
    print("U(1) symmetry satisfied!!")
```

problem #3

Find the ground state of the Hamiltonian by optimizing parameterized quantum circuits with the U(1) symmetry.

second quantized

1 … electron O

0 … electron X

Electron cannot occupied more than one orbital.
One orbital has or does not have an electron.

How to place the electrons

: There is an empty orbital, which orbital choose?

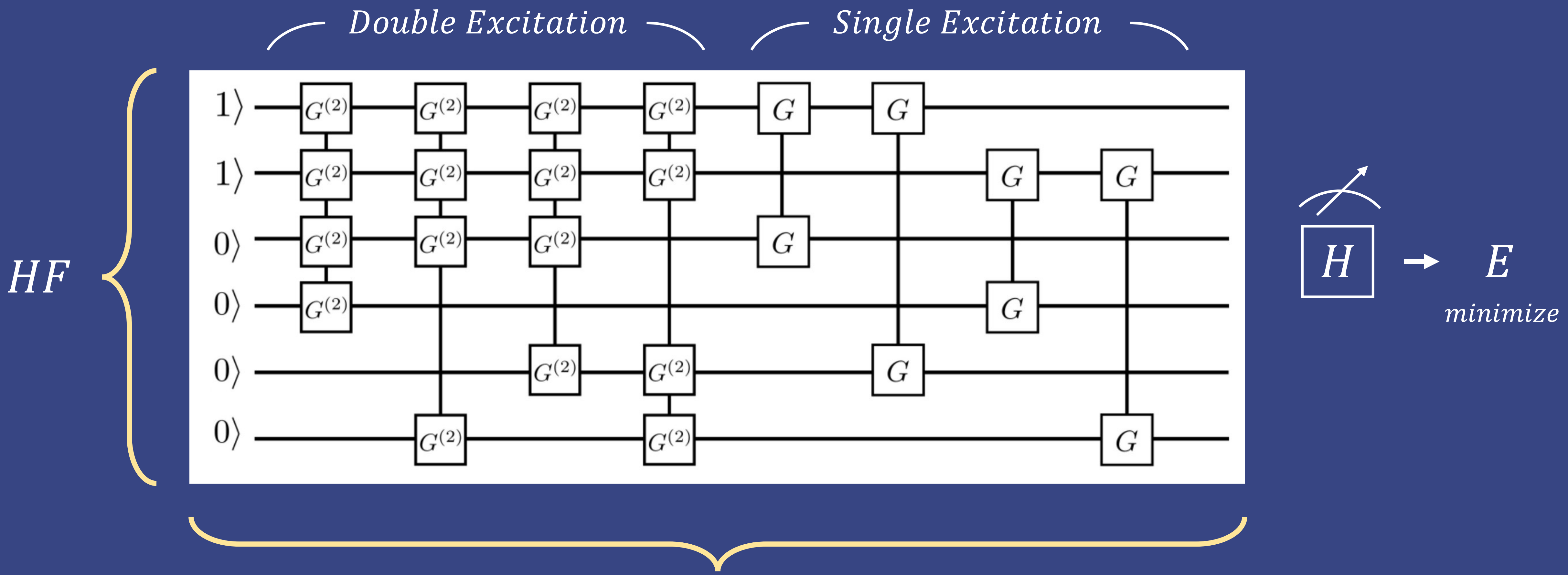
The number of electrons must be preserved

U1 symmetry

Set of operators that preserve the number of '1' in qubit

The number of qubits is naturally preserved. The search space is greatly reduced in Hilbert space.

problem #3



$ansatz = qml.AllSingleDoubles$

main/N2_16_U(1).ipynb </>

problem #3

Find the ground state of the Hamiltonian by optimizing parameterized quantum circuits with the U(1) symmetry.

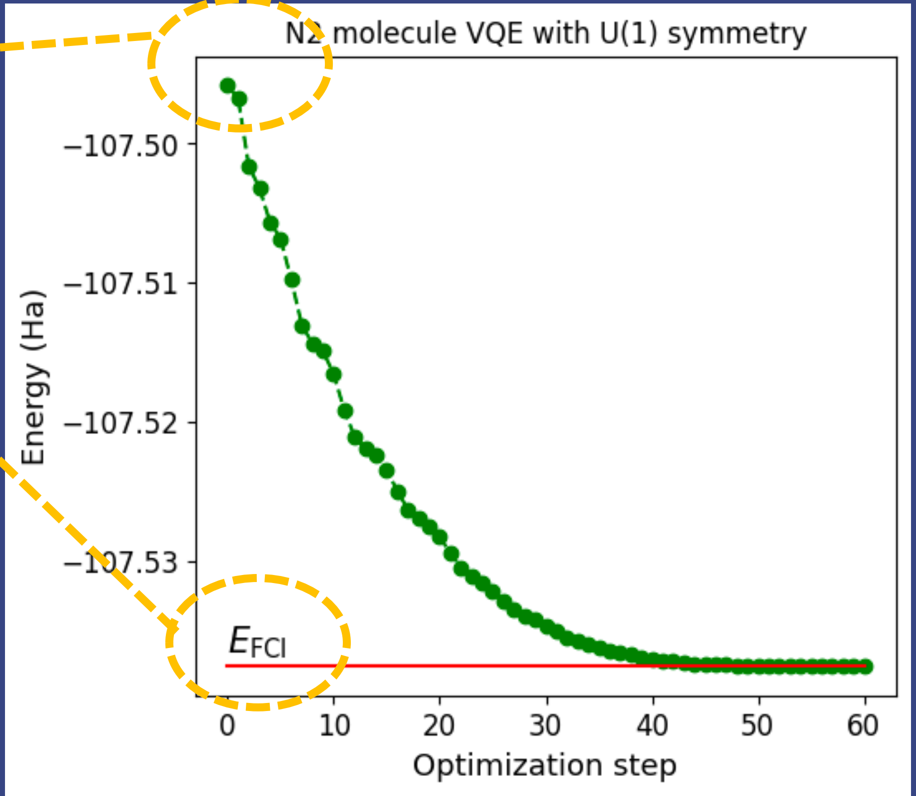
E_{HF}

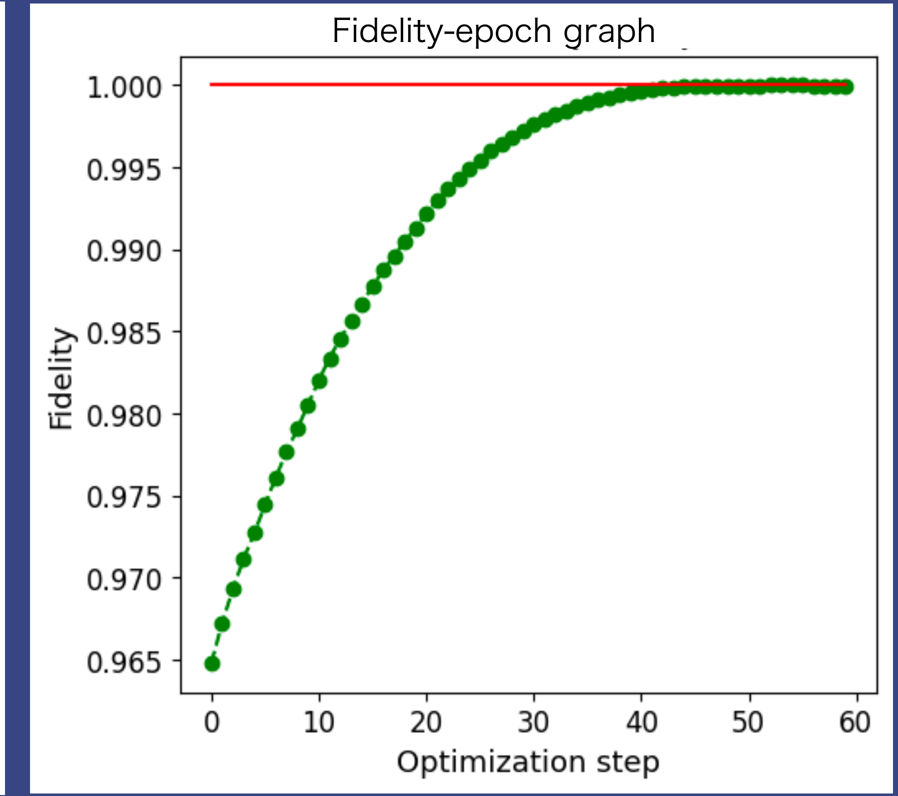
$|1111111111111100\rangle$

ground state

combination of these three is lower than

$Amplitude\ of\ state\ |1111111111111100>$
$=\ 0.9628454299159072$
$Amplitude\ of\ state\ |1111111111001111>$
$=\ 0.03337755511475772$
$Amplitude\ of\ state\ |1111110011111111>$
$=\ 0.00250472260884314$
… …





open question

1) Can you do this for the SU(2) symmetry, i.e., find a two-qubit gate W such that] for all U ∈ SU(2)?

$$\left(U^{\otimes 2}\right)^{\dagger} W U^{\otimes} = W$$

$$SU(2)$$

2 x 2

$$U^{\dagger}U = I \quad (imply \,|\det U| = 1)$$

$$\det U = 1$$

open question

$$U = \begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix} \quad \longrightarrow \quad U^{\otimes 2} = \begin{bmatrix} \alpha^2 & -\overline{\alpha\beta} & -\overline{\alpha\beta} & \overline{\beta^2} \\ \alpha\beta & |\alpha|^2 & -|\beta|^2 & -\overline{\alpha\beta} \\ \alpha\beta & -|\beta|^2 & |\alpha|^2 & -\overline{\alpha\beta} \\ \beta^2 & \bar{\alpha}\beta & \bar{\alpha}\beta & \bar{\alpha}^2 \end{bmatrix}$$

$$\left(U^{\otimes 2}\right)^{\dagger} W U^{\otimes 2} = W \quad \Rightarrow \quad W U^{\otimes 2} = U^{\otimes 2} W$$

$$W^{\dagger} W = I$$

unitary condition

open question

$$SU(2) = \left\{ \begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix} \middle| \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1 | \right\}$$

$$W = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}$$

$$U^{\otimes 2} = \begin{bmatrix} \alpha^2 & -\overline{\alpha\beta} & -\overline{\alpha\beta} & \overline{\beta^2} \\ \alpha\beta & |\alpha|^2 & -|\beta|^2 & -\overline{\alpha\beta} \\ \alpha\beta & -|\beta|^2 & |\alpha|^2 & -\overline{\alpha\beta} \\ \beta^2 & \overline{\alpha}\beta & \overline{\alpha}\beta & \overline{\alpha}^2 \end{bmatrix}$$

$$WU = UW$$

System of linear equations

$(1,1)$: $a\alpha^2 + (b+c)\alpha\beta + d\beta^2 = a\alpha^2 - (e+i)\alpha\bar{\beta} + m\bar{\beta}^2$

$\therefore \boldsymbol{b + c = 0}, \quad \boldsymbol{e + i = 0}, \quad \boldsymbol{d = m = 0}$

$(4,4)$: $-(n+o)\overline{\alpha\beta} + p\bar{\alpha}^2 = (h+l)\bar{\alpha}\beta + p\bar{\alpha}^2 \quad \therefore \boldsymbol{n + o = o}$

$(1,2)$: $-a\alpha\bar{\beta} + b|\alpha|^2 + b|\beta|^2 = b\alpha^2 - f\alpha\bar{\beta} - j\alpha\bar{\beta} + \mathrm{n}|\beta|^2$

$b = b\alpha^2 + n\bar{\beta}^2 \therefore \boldsymbol{b = n = 0}$

$(2,1)$: $e\alpha^2 + (f+g)\alpha\beta + h\beta^2 = a\alpha\beta + 3$

$\therefore \boldsymbol{f + g = a}, \quad \boldsymbol{h = 0}, \quad \boldsymbol{e = 0}, \quad \boldsymbol{f + g = f + j = a}$

$(2,2)$: $f|\alpha|^2 - g|\beta|^2 = f|\alpha|^2 - j|\beta|^2 \quad \therefore \boldsymbol{g = j}$

$(3,3)$: $-g|\beta|^2 + k|\alpha|^2 = -g|\beta|^2 + k|\alpha|^2$

$(3,4)$: $-(g+k)\overline{\alpha\beta} = p(-\overline{\alpha\beta}) \therefore \boldsymbol{p = g + k}$

$(4,1)$: $p\beta^2 = a\beta^2 \rightarrow a = p$

$\therefore \boldsymbol{f + g = a}, \quad \boldsymbol{g + k = a}, \quad \boldsymbol{f = k}$

open question

$$\begin{bmatrix} a & b & -b & 0 \\ e & f & g & h \\ -e & j & k & -h \\ 0 & n & -n & p \end{bmatrix}\begin{bmatrix} \alpha^2 & -\alpha\bar{\beta} & -\alpha\bar{\beta} & \bar{\beta}^2 \\ \alpha\beta & |\alpha^2| & -|\beta^2| & -\overline{\alpha\beta} \\ \alpha\beta & -|\beta^2| & |\alpha^2| & -\overline{\alpha\beta} \\ \beta^2 & \bar{\alpha}\,\beta & \bar{\alpha}\,\beta & \bar{\alpha}^2 \end{bmatrix} \quad \text{vs} \quad \begin{bmatrix} \alpha^2 & -\alpha\bar{\beta} & -\alpha\bar{\beta} & \bar{\beta}^2 \\ \alpha\beta & |\alpha|^2 & -|\beta|^2 & -\overline{\alpha\beta} \\ \alpha\beta & -|\beta|^2 & |\alpha|^2 & -\overline{\alpha\beta} \\ \beta^2 & \bar{\alpha}\beta & \bar{\alpha}\beta & \bar{\alpha}^2 \end{bmatrix}\begin{bmatrix} a & b & -b & 0 \\ e & f & g & h \\ -e & j & k & -h \\ 0 & n & -n & p \end{bmatrix}$$

$$\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & f & g & 0 \\ 0 & g & f & 0 \\ 0 & 0 & 0 & p \end{bmatrix}\begin{bmatrix} \alpha^2 & -\alpha\bar{\beta} & -\alpha\bar{\beta} & \bar{\beta}^2 \\ \alpha\beta & |\alpha^2| & -|\beta^2| & -\overline{\alpha\beta} \\ \alpha\beta & -|\beta^2| & |\alpha^2| & -\overline{\alpha\beta} \\ \beta^2 & \bar{\alpha}\,\beta & \bar{\alpha}\,\beta & \bar{\alpha}^2 \end{bmatrix} \quad \text{vs} \quad \begin{bmatrix} \alpha^2 & -\alpha\bar{\beta} & -\alpha\bar{\beta} & \bar{\beta}^2 \\ \alpha\beta & |\alpha|^2 & -|\beta|^2 & -\overline{\alpha\beta} \\ \alpha\beta & -|\beta|^2 & |\alpha|^2 & -\overline{\alpha\beta} \\ \beta^2 & \bar{\alpha}\beta & \bar{\alpha}\beta & \bar{\alpha}^2 \end{bmatrix}\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & f & g & 0 \\ 0 & g & f & 0 \\ 0 & 0 & 0 & a \end{bmatrix}$$

$\Rightarrow$ we can express with only 3 variables **(a, f, g)**

$$\begin{bmatrix} \bar{a} & 0 & 0 & 0 \\ 0 & \bar{f} & \bar{g} & 0 \\ 0 & \bar{g} & \bar{f} & 0 \\ 0 & 0 & 0 & \bar{a} \end{bmatrix}\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & f & g & 0 \\ 0 & g & f & 0 \\ 0 & 0 & 0 & a \end{bmatrix} \qquad \therefore |\boldsymbol{a}| = \mathbf{1}, \quad |\boldsymbol{f}|^2 + |\boldsymbol{g}|^2 = \mathbf{1}, \quad \boldsymbol{f} + \boldsymbol{g} = \boldsymbol{a}, \quad \bar{\boldsymbol{f}}\boldsymbol{g} + \boldsymbol{f}\bar{\boldsymbol{g}} = \mathbf{0}$$

open question

$$\therefore\ |a| = 1, \qquad |f|^2 + |g|^2 = 1, \qquad f + g = a, \qquad \bar{f}g + f\bar{g} = 0$$

$$SWAP = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$SWAP$ and I are include in $SWAP^{\alpha}$

$$SWAP^{\alpha} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+e^{i\pi\alpha}}{2} & \frac{1-e^{i\pi\alpha}}{2} & 0 \\ 0 & \frac{1-e^{i\pi\alpha}}{2} & \frac{1+e^{i\pi\alpha}}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$a = 0 \quad \rightarrow \quad I$$

$$a = 1 \quad \rightarrow \quad SWAP$$

open question

$$IsingXY(\phi) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \cos\frac{\phi}{2} & i\sin\frac{\phi}{2} & 0 \\ 0 & i\sin\frac{\phi}{2} & \cos\frac{\phi}{2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$IsingZZ(-\frac{\phi}{2}) = \begin{bmatrix} e^{i\phi/4} & 0 & 0 & 0 \\ 0 & e^{-i\phi/4} & 0 & 0 \\ 0 & 0 & e^{-i\phi/4} & 0 \\ 0 & 0 & 0 & e^{i\phi/4} \end{bmatrix}$$

$$IsingXY\left(-\frac{\phi}{2}\right) IsingZZ(\phi) = \begin{bmatrix} \boxed{e^{i\phi/4}} & 0 & 0 & 0 \\ 0 & e^{-i\phi/4}\cos\frac{\phi}{2} & ie^{-i\phi/4}\sin\frac{\phi}{2} & 0 \\ 0 & ie^{-i\phi/4}\sin\frac{\phi}{2} & e^{-i\phi/4}\cos\frac{\phi}{2} & 0 \\ 0 & 0 & 0 & e^{i\phi/4} \end{bmatrix}$$

$\therefore |a| = 1, \quad |f|^2 + |g|^2 = 1, \quad f + g = a, \quad \bar{f}g + f\bar{g} = 0$

$$\boldsymbol{e^{-i\phi/4}\left(\cos\frac{\phi}{2} + i\sin\frac{\phi}{2}\right) = e^{-i\phi/4}\, e^{i\phi/2} = e^{i\phi/4}}$$

open question

2) Can we also find <u>three-qubit</u> gates with this property?

three-qubit states?

➡ Found three!

assume there are more, but can't categorize them

$3\ input\ SWAP$ $\rightarrow$ $3\ input\ SWAP^{\alpha}$ $\rightarrow$ $IsingXY\left(-\frac{\phi}{2}\right) IsingZZ(\phi)$

$2\ set\ of$

$2\ input\ SWAP$

$(2 \leftrightarrow 3) \rightarrow (1 \leftrightarrow 2)$

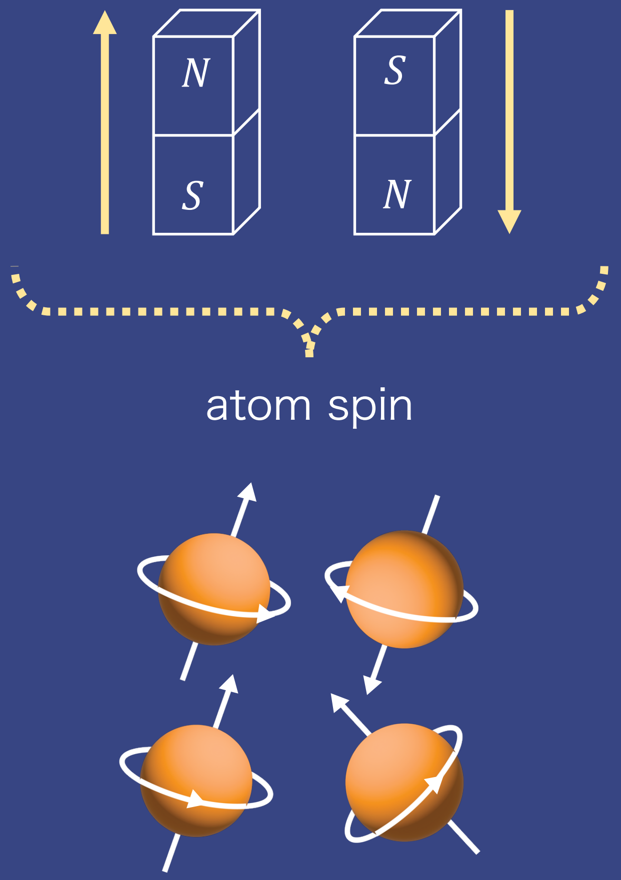
$$\left\{ \begin{array}{c} |\, i\ j\ k\, \rangle \\ |\, k\ i\ j\, \rangle \end{array} \right.$$

3 qubit $SU(2)$ symmetry

3 qubit $SU(3)$ symmetry

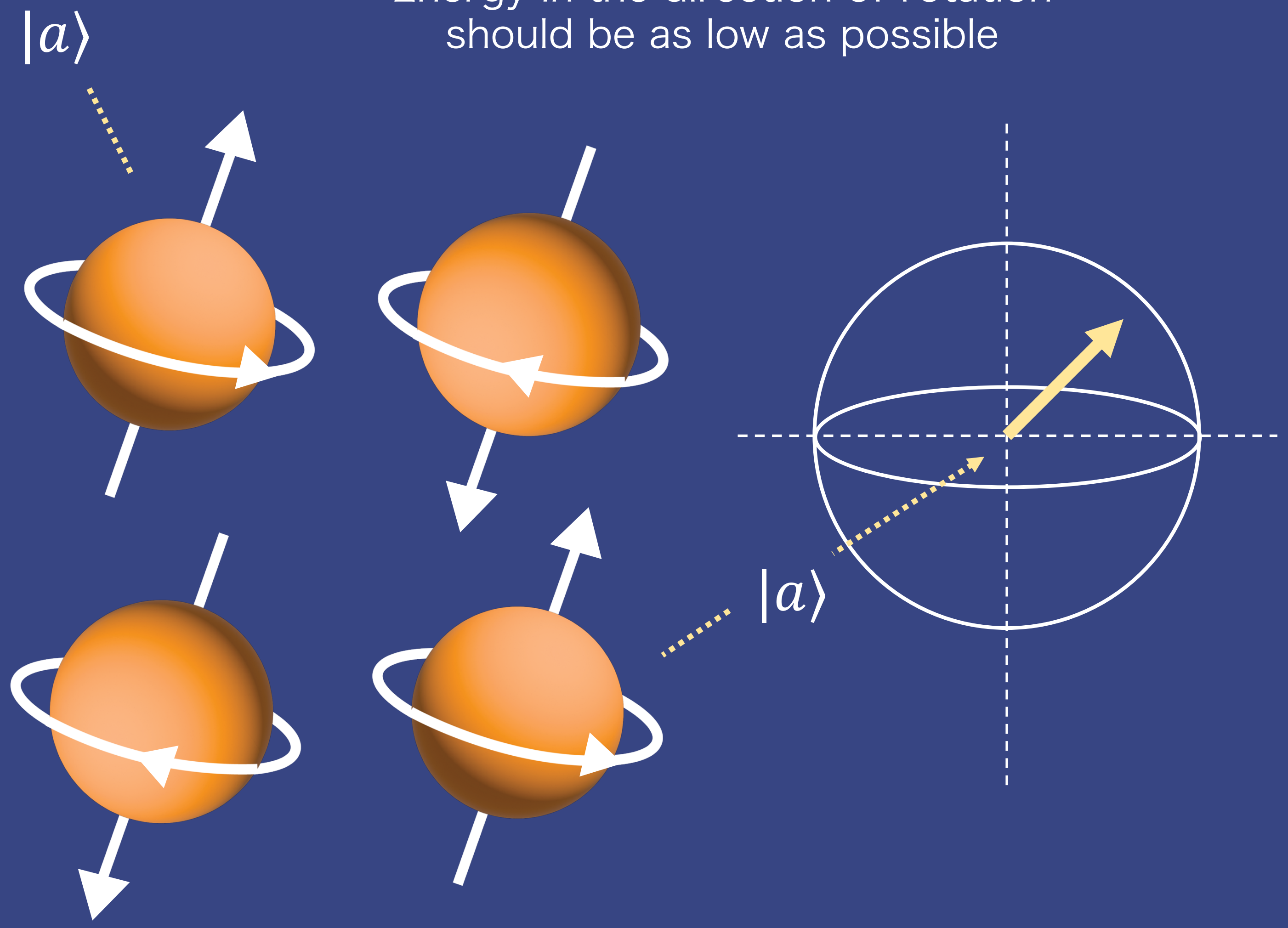
open question

Optimizing Hamiltonian



qubit has direction

Energy in the direction of rotation
should be as low as possible



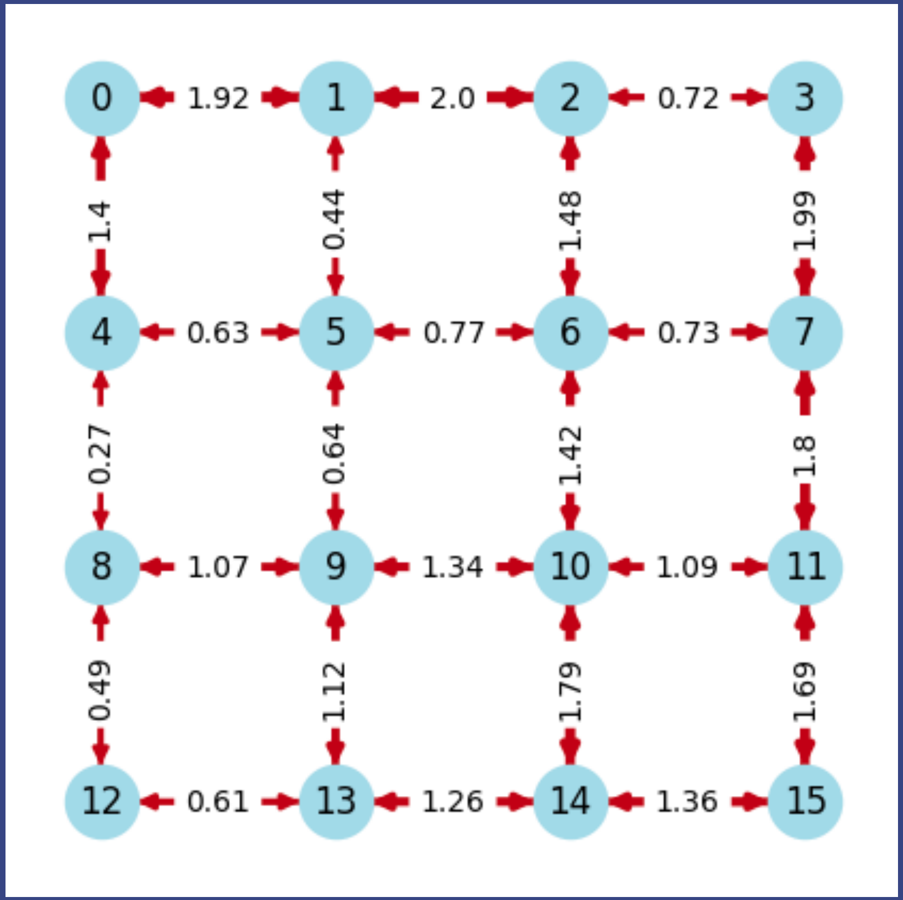
open question

3) Devise a problem with the SU(2) symmetry and solve it using those gates.

we choose…

2D Heisenberg Model

$$H = \sum_{i<j} J_{ij}\left(X_i X_j + Y_i Y_j + Z_i Z_j\right)$$



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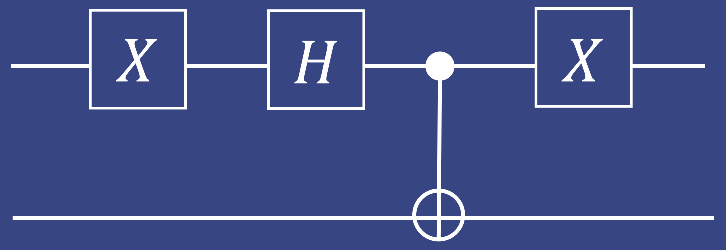
open question

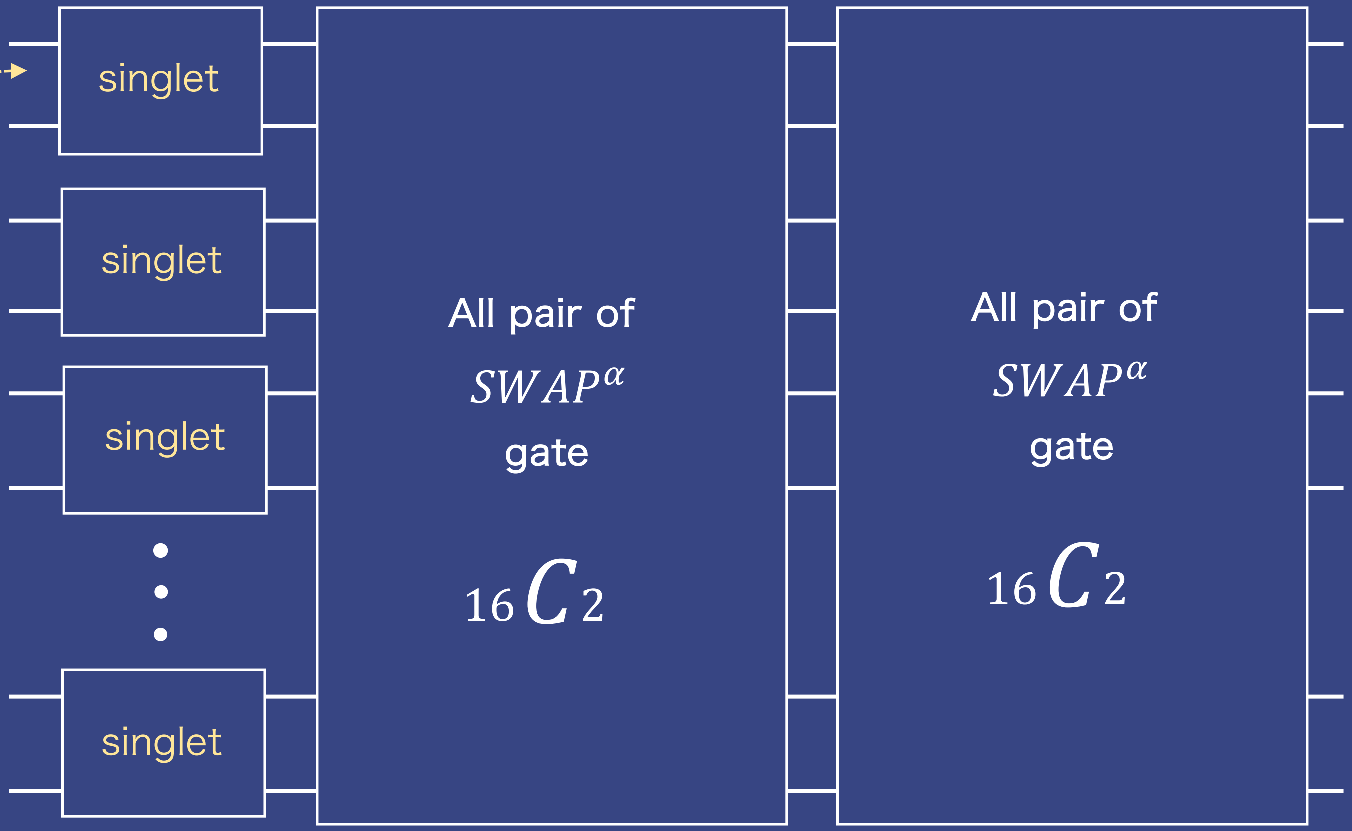
Exploit SU(2) symmetry!

Why singlet?



< singlet state >

the total spin of the Heisenberg model ground state is zero



< our SU(2) symmetry ansatz >

main/Heisenberg_2D_SU(2).ipynb </>
main/Heisenberg_2D_U(1).ipynb </>
main/Heisenberg_2D_general.ipynb </>

open question